PRESENTER NOTES — WEEK 5: LOCKING THE LINK

GROUP 9 · OFF-GRID VEHICLE TRACKER · REV E · 18 SHEETS · 2026-09-25
Full solo-presenter scripts. Sheets 17–18 are reference sheets — point to them once, then use for Q&A.

SHEET 01/18

Cover — Locking the Link

Opening: last review published the plan; this one reports the milestone. Both radios now hold a verified two-way link on our private channel. But the headline is not 'it works' - it is WHY it did not work for a day with byte-identical configurations, and what that taught us about how LoRa mesh radios actually lock onto each other. Every claim on these sheets was measured on our own hardware this week, most of it twice.

SHEET 02/18

Two Radios, One Contract: What Must Match

Frame the sheet as a contract: LoRa has no negotiation, no handshake at the PHY - both sides must independently arrive at identical radio parameters from their own stored config. The table is ordered like the OSI stack: frequency first (can you even receive energy), then modulation (can you demodulate chirps), then channel crypto (can you read the payload), then per-node identity (can you address a specific radio). Our failure lived in row one - the least visible layer, because every tool reports the CONFIG, not the synthesizer frequency. Segue: so where does that frequency actually come from? Next sheet is the math.

SHEET 03/18

The Channel Name IS the Frequency

Walk the equation left to right: start of band, plus half a bandwidth so the slot is centered, plus slot index times bandwidth. The slot index is a plain djb2 string hash of the channel name modulo the slot count - show the six-line loop, this is the entire algorithm. Verify it live if asked: LongFast hashes to 19, which is 906.875 - the published US default, so the math is calibrated against a known constant. Group9 hashes to 43 = 912.875. Point at the band figure: the two markers are 6 MHz apart. A radio on one marker receives effectively nothing from the other - at 250 kHz bandwidth, 6 MHz off-tune is total isolation. That is the trap: the name looks like a label, but it is a tuning word.

SHEET 04/18

Root Cause: Same Key, Different Slot

The 2x2 matrix is the whole diagnosis space. Top-left is health. Top-right - same slot, wrong key - still updates 'last heard' counters because packet headers are cleartext; only payloads drop. Bottom-left is OUR cell and the cruelest: byte-identical keys, verified configs, zero RF coupling, because the radios were 6 MHz apart. Every software check reads back the intended config while the synthesizer sits on the old slot until a clean save+reboot. Method that isolated it: A/B regression - both radios moved to the known-good default channel, link immediately proven with round-trip traceroute; both moved to Group9 with the OLED frequency verified at 912.875, link proven again. Two variables, tested one at a time.

SHEET 05/18

Locking On: From First Packet to Addressable Peer

This is the 'locking' the title promises, and the key point is what is NOT here: no connection, no association, no session like WiFi or Bluetooth. Acquisition is one-directional bookkeeping - I heard you, I filed you, I kept your public key. Steps 4 and 5 split the traffic classes: broadcasts need only the shared channel key (works from the first packet), direct messages are individually encrypted to a node's public key and fail with the exact NAKs shown until NodeInfo has crossed. Both NAKs in the code box are our real captures. The second one is a diagnostic gem: an error REPLY that arrives by radio proves the RF path while the application layer is still refusing - we used it as a link test when nothing else would talk.

SHEET 06/18

Proof of Lock: Round-Trip Traceroute

Why traceroute and not a text message: a text proves one direction and needs a human at the far screen; traceroute is machine-checkable, round-trip, and reports per-hop SNR from the receiving demodulator. Explain the margin bar: LoRa's spreading gain means SF11 decodes down to 17.5 dB BELOW the noise floor - that is the negative-SNR magic of chirp spread spectrum. At +6 dB measured, we hold 23.5 dB of margin; every 6 dB of margin is roughly a doubling of free-space distance, which is why the same numbers that look boring on a desk translate to kilometers outdoors. This measurement is the baseline for the W8-W9 range characterization: same command, growing distance, log SNR vs range.

SHEET 07/18

\u201cOnline\u201d Is an Equation, Not a Connection

The photo is the sheet: signal GOOD and 'heard 72 days ago' in the same breath - both true, because they answer different questions. Signal quality is measured per packet; 'online' is arithmetic on a stored timestamp against the local clock. Three ways that arithmetic lied to us in one afternoon: a node with no GPS fix rebroadcast its STORED position carrying the old fix time, dragging the peer's last-heard back to July; a reboot reloaded month-old records from flash because the node DB only persists on its own schedule; and after a clock-less reboot the same stale records counted as ONLINE - the screen said 17 online from a two-month-old node list. Same inequality, three failure modes, all timestamp hygiene. None of this can occur in a vehicle with sky view - fresh fixes stamp fresh times continuously.

SHEET 08/18

The Heartbeat: Designed Traffic Cadence

This table IS the protocol design of the product - everything else is plumbing. Two rates matter for the demo: 10 s / 20 m while moving (the tracking experience) and the 300 s telemetry floor (the honesty guarantee). The smart broadcast is an AND gate: at highway speed the 20 m distance gate is crossed in under a second, so the 10 s timer dominates; walking pace flips it. Defend the 300 s choice with the two arguments in the bullets: it is 6x denser than the failure window it patches, and 60x cheaper than the channel budget cares about. The rule in the last bullet is worth stating out loud to the sponsor: no packet flows on this channel without a named failure it prevents.

SHEET 09/18

Airtime: Every Packet Costs Half a Second

Time-on-air is the currency of everything on this channel. Walk the equation: symbol time is two-to-the-SF over bandwidth - at SF11/250k that is 8.19 ms, so even a lean position packet spends around half a second on air. Then the observation that surprised us on the bench: channel utilization ran 9-13% with two IDLE radios, because a flood mesh relays by default and with exactly two nodes every relay is a perfect waste - there is no third radio for the echo to reach. The budget box is the punchline: at drive-test rates the default behavior lands on ~20%, brushing the 25% duty throttle that silently suppresses position broadcasts - the product's core packet - precisely when both vehicles move. CLIENT_MUTE buys the headroom back for free, with the revert condition documented.

SHEET 10/18

Link Budget: From Bench Meters to Field Kilometers

Read the graph left to right: received power falls 6 dB per doubling of distance; the red dashed line is the SX1262's decode floor at our modulation. The gap between the blue curve and the red line IS the link margin - at bench range it is enormous (about 100 dB), which is why nothing about a desk test predicts range. The dotted green line moves the requirement 20 dB up to absorb real-world fading and obstruction; where the blue curve meets THAT line is the honest few-kilometer design expectation, matching our 3-5 km target for the v2v use case. The free-space crossing near 100 km is a math fact, not a claim - treat it as an upper bound that terrain will spend. Range test protocol: fixed SF11, log traceroute SNR vs GPS distance, plot against this exact chart.

SHEET 11/18

The Display Needs Armor: Case Selection

PRESENTER PREP - Sheet 11 (Josiah, ~90 s). Open with the incident: our first display arrived cracked and dead - that is not an abstract risk, it already cost us a replacement request to the sponsor. State the requirement set as engineering constraints, not wishes: every port and button the firmware needs must stay reachable, and the case must add zero RF attenuation around the BLE antenna - plastic, not metal, which is why 3D printing fits. Explain the selection: an existing open model by the maker AWSW on Printables (model number 1610004) matches the exact panel; the license is Creative Commons Attribution-NonCommercial 4.0, which permits our academic use with credit - hence the attribution printed on this sheet. Vocabulary to be ready for: RLCD = reflective LCD, the sunlight-readable panel type; 18650 = the cylindrical lithium cell size the board's holder takes; M2.5x6 = metric screw, 2.5 mm thread diameter, 6 mm length; BLE = Bluetooth Low Energy, the radio link between display and Heltec. Question you may get: 'why not design our own?' - answer: zero schedule cost beats originality; we validate fit first, and we can remix later because the license allows derivatives.

SHEET 12/18

Print Engineering: Parts, Settings, Slicing

PRESENTER PREP - Sheet 12 (Josiah, ~2 min). Walk the parts table first - seven pieces, and the reason the back is TWO pieces is functional: the split leaves the 18650 battery bay and speaker reachable without disassembly. Then translate every slicer term on the sheet, because the room will not all be printer people: 3MF is the modern print file format (geometry PLUS settings, unlike bare STL); layer height 0.2 mm means each deposited slice is a fifth of a millimeter; supports are sacrificial scaffolding for overhangs and 'organic' means tree-like trunks that touch less surface, so less scarring; fuzzy skin is deliberate nozzle jitter for texture; a brim is a one-layer skirt that fights warping - this model doesn't need one; sequential printing finishes each small part before starting the next so a mid-plate failure cannot wreck everything. Material call: PETG over PLA for a vehicle - a dashboard in Texas sun exceeds PLA's ~60 C glass transition; PETG holds to ~80 C. Point at the bed photo: that is the author's actual plate on a Prusa MK3 - organic supports visible inside the frame. Our plan box: one overnight run, about a dollar of filament, timed to the replacement panel's arrival.

SHEET 13/18

Assembly, Access, and Vehicle Fit

PRESENTER PREP - Sheet 13 (Josiah, ~90 s). Assembly is four steps and one warning: the M2.5 holes are IN the display's own back, so screw length is a hard limit - the author flags that overlong screws press the glass; 6 mm, no substitutes. Explain what stays reachable and why it matters operationally: USB-C is how we flash and read logs, the SD slot is future data logging, the three buttons are BOOT (bootloader entry for flashing), RESET, and the user button. RF point for the sponsor: the enclosure is thermoplastic, effectively transparent at 2.4 GHz, so the BLE link to the Heltec is untouched - but we still verify by comparing RSSI (received signal strength indicator) cased vs bare, budgeting 2 dB as the pass line. Thermal point: this is a REFLECTIVE LCD - it uses ambient light, no backlight, so the panel adds almost no heat; the ESP32-S3 under load is below a watt, and the port cutouts provide the airflow. Close on the acceptance box - four measurable checks, same discipline as the radio sheets: nothing counts as done until each line is ticked on the bench.

SHEET 14/18

Next Milestone: Two-Vehicle Drive Test

PRESENTER PREP - Sheet 14 (Cannon, ~2 min). This is the test we could not run until this week - every prerequisite in the pink box was closed on September 25. Walk the five phases in order and give each its purpose: Static A calibrates what 'good' looks like at trivial range; Convoy is the actual product experience - the display's partner age should tick 10-15 s continuously; Split turns distance into data - we drive apart in steps and log SNR (signal-to-noise ratio) against GPS distance, which becomes the measured curve to overlay on the link-budget prediction; Rejoin measures how fast the pair re-locks after real silence - our acceptance is under a minute; Stop test exercises the canned-message chain end to end. Emphasize the discipline line: metrics are declared BEFORE the drive, so success cannot be redefined afterward - p95 means the 95th percentile, i.e., all but the worst 5% of samples must beat 15 seconds. Instrumentation point: the radios themselves are the loggers; node databases are pulled over USB between phases, no laptop rigs in cars. Hand-off: the same logs seed the Week 8-9 formal range characterization.

SHEET 15/18

The Bench Record: Four Screens, One Story

Four photos, chronological. One: the as-received node table - eighteen nodes from the sponsor's summer mesh, all months stale, the raw material of every counter lie on sheet 07. Two: a text landing during the debug session - the moment we knew the RF layer was healthy and the fault was bookkeeping. Three: the same table after the cleanup - peer heard one minute ago, and it stays that way. Four: the home screen we now read as a dashboard - uptime catches silent reboots, voltage catches the charging state, channel utilization catches airtime regressions. The discipline banner is the transferable result: one variable per trial, read-back plus OLED verification, machine-checkable probes, and treat every NAK as data.

SHEET 16/18

As-Built Configuration and References

Close on the as-built table - this is the contract the drive test inherits, every row verified twice (read-back plus OLED). References: [1] backs every sensitivity and SNR number quoted; [2] and [6] together explain the root-cause sheets - name-derived frequency and layered encryption; [3] and [4] back the airtime sheet; [5] the heartbeat; [7] the time-on-air equation; [8] the regulatory envelope for +30 dBm in this band. Next milestones on the board: two-vehicle drive test (in progress, 20%), then range characterization W8-W9 plotted against sheet 10's predicted curve.

SHEET 17/18

Nomenclature I: Acronyms & Abbreviations

PRESENTER PREP - Sheet 17 (whoever fields questions; do not read aloud). This sheet exists so no acronym in the deck is unexplained - point the audience to it once, early ('everything shorthand is decoded on sheet 17'), then leave it for Q&A. Highest-value entries to know cold if asked: SF (spreading factor) - 2 to the SF chips encode each symbol, which is why SF11 is slow but decodes below the noise; dB vs dBm vs dBi - a dB is always a RATIO, dBm anchors it to one milliwatt of power, dBi anchors antenna gain to a perfect point radiator; PSK vs PKC - the channel key everyone shares versus the per-node keypair used only for direct messages - our week's NAKs came from the second one; djb2 - a 1990s string hash, five lines of code, that quietly doubles as Meshtastic's frequency planner. RLCD deserves one sentence: reflective LCD uses ambient light instead of a backlight - that is why it is readable in direct sun on a dashboard and why the case has no thermal problem.

SHEET 18/18

Nomenclature II: Every Formula, Symbol by Symbol

PRESENTER PREP - Sheet 18 (reference; walk it only if asked). Each quadrant re-prints one of the deck's four equations and decodes it term by term. Frequency slot: read it as 'start of band, plus half a channel to sit centered, plus hash-of-the-name channels up' - the whole root cause of our silent week lives in that middle term. Liveness: the double arrow means equivalence - 'online' is nothing but this comparison; epoch means seconds counted from January 1st 1970, the universal computer clock. Time on air: the only non-obvious constant is 4.25 - a fixed fraction of the preamble the LoRa standard adds beyond the counted symbols; everything else is symbol counting times 8.19 ms. Link budget: minus 147.55 looks magic but is just the unit bookkeeping of meters and hertz in free-space path loss; the physical content is '+6 dB of loss per doubling of distance'. If someone asks why sensitivity is negative: dBm is logarithmic, so -131 dBm is a tenth of a millionth of a millionth of a milliwatt - the radio decodes signals that small because SF11 spreads each bit across 2048 chips.